



NEWS79 ***December 2016***

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FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Contents

Editorial

Announcements

Total number and an updated listing of flea species of the world (continued)

Featured research

Fleas in art and history

Francis Bacon, *Novum Organum* (1620)

Domencio Gargiulo, Piazza Mercatello in Naples during the Plague of 1656 (circa 1675)

Directory of Siphonapterists

Siphonaptera Literature 2016

Editorial

Dear *Flea News* Reader,

In regard to fleas, it seems there is a need for taxonomic revision at several levels. Order Siphonaptera (fleas) appear nested within a family (Boreidae) of order Mecoptera (scorpionflies) (Whiting 2002, Whiting et al. 2008), though their preliminary results need replication.

Orders. Mecopterans may need elevation to a level above order. Mecopterans are an ancient and diverse order possibly important as pollinators of plants such as tropical evergreens of *Gnetum*, and extinct seed ferns and conifers. Further, Mecopterans have both aquatic and terrestrial groups. Mecopterans are hypothesized to be ancestral to Trichoptera (caddisflies), Lepidoptera (butterflies and moths), Diptera (flies).

Boreidae could be elevated to a level above family, or Siphonaptera could be demoted to suborder, or even family, level. Siphonaptera placed as a family of Mecoptera would resemble the taxonomic arrangement of Culicidae (mosquito) which are also specialized blood-suckers in Diptera. Interestingly, there are similar numbers of taxa in fleas (2500 species in 240 genera) and mosquitoes (3500 species in 41 genera).

Families. Several families of fleas (Ctenophthalmidae, Hystrichopsyllidae, and Leptopsyllidae) appear paraphyletic (that is, similar species are not grouped together) (Whiting et al. 2008).

Species. There are also questions on flea species. Hybridization seems common among fleas. There have been cladistic (grouping taxa by distinctive, shared characteristics that are presumably derived from near ancestors but are not present in distant ancestors) and phylogenetic studies, but there has been no classical genetics work on fleas, so how many species are invalid (Lewis 1996, Zurita et al. 2016)?

Subspecies. There are more subspecies per species of flea than perhaps any other order (31% of named flea taxa are subspecies, Lewis 1996). Is the high number of subspecies a real biological phenomenon of fleas, perhaps due to repeated founder effects or peculiar population genetics (Bossard et al. 1998), or are many subspecies simply an artifact of taxonomic description? Although there is a lot of morphological variation among flea species, the evolutionary and ecological significance of these variations is often obscure because we know little about behavior and life cycles.

If revision at multiple taxonomic levels is needed, who will revise?

Yours in fleas,

R.L. Bossard, Editor, *Flea News*

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Announcements

Total Number, and an Updated Listing, of Flea Species of the World (continued from the preceding *Flea News*)

Mike Hastriter and I continue to update a list of all flea species in the world on the *Flea News* network (Entomological Society of America website). I thank everyone who contributed. Please send me any corrections.

Many thanks to M.W. Hastriter who created the spreadsheet file for the current list that was from the following lists, and to *Flea News* readers and editors. The spreadsheet lists families, subfamilies, and synonyms for flea species.

Hastriter, M.W. (2016). Pers. comm., *Flea News* 77.

Lewis, R. E., & Lewis, J. H. (1989). A catalogue of invalid or questionable genus-group and species-group names in the Siphonaptera (Insecta). Königstein: Koltz Scientific Books

Lewis, R.E. 2009. Siphonaptera (parts I-III), 16th ed. Unpublished list, 1 November 2009.

Medvedev, S. 2016. <http://www.zin.ru/Animalia/Siphonaptera/taxfind2>

R.L. Bossard
Editor, *Flea News*

Nancy Hinkle asks: does the *Nosopsyllus fasciatus* male have a groove in its head like *Ctenocephalides* does? How common is it throughout Siphonaptera for males to have the cephalic groove?

**

Congratulations to Miss Theresa M. Howard upon her retirement from the Natural History Museum in London where she managed Siphonaptera and Diptera Collections. The Flea News editors thank her for supporting Flea News for many years.

The new contact for Fleas at the NHM will be: Dr. Erica McAlister (see directory in this Flea News).

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Doctoral Fellowship Position Available

Director Marcela Lareschi is looking for interested applicants for a position of a Doctoral Fellowship at Facultad de Ciencias Naturales y Museo, Universidad Nacional de La Plata, Argentina, with support by FONCyT (www.fcnym.unlp.edu.ar/ ; <http://www.agencia.mincyt.gob.ar/frontend/agencia/fondo/foncyt>).

The project will involve Flea and Mite Parasites of Sigmodontine Rodents (Cricetidae) in the Neotropical Region of Argentina. Factors and Processes that Modulate their Distribution.

The place of work will be the Ectoparasite Laboratory, CEPAVE (www.cepave.edu.ar).

Anticipated start date: March 1st, 2017

Duration of the Scholarship: 3 years.

Requirements to Apply: Graduated in Biology, Veterinary or related sciences. Preferably with knowledge of bioinformatics and molecular biology. With basic knowledge of entomology. Ability to work in a team and learn laboratory techniques to develop their research work. Knowledge of English.

Submission of applications up to January 2017.

If interested, please submit a Curriculum Vitae and a short letter of motivation to join the working group, detailing relevant previous experiences related to the requirements and contact details of two people who can provide references about the applicant. Subsequently, a personal or Skype interview can be made.

Contact: Marcela Lareschi mldareschi@cepave.edu.ar mldareschi@yahoo.com.ar

**

Pet Ectoparasite Presentations for 61st LIWC

The 61st Livestock Insect Workers Conference will be held June 24-28, 2017, in Savannah, GA.

A special session this year will include ectoparasites, fleas, ticks, lice, mites, mosquitoes, and any other arthropod pests, of small animals, including companion animals (dogs, cats, horses, and other pets). We welcome submissions for research presentations on topics that might otherwise be presented at the International Symposium on Ectoparasites of Pets. Please contact Nancy Hinkle (NHinkle@uga.edu) to be added to the mailing list.

Featured Research

New Fleas 2016

M. Fernanda López Berrizbeitia, Michael W. Hastriter and M. Mónica Díaz. 2016. A New Flea Species of the Genus *Cleopsylla* (Siphonaptera: Stephanocircidae) from Northwestern Argentina. *The Journal of Parasitology* 102(5).

[*Cleopsylla barquezi* Berrizbeitia, Hastriter, and Díaz 2016]

Hastriter, M. W. (2016). Fleas (Siphonaptera: Pygiopsyllomorpha) of Papua New Guinea and Papua province (Indonesia). Part IV. *Rectidigitus* (Stivaliidae: Stivaliinae), description of four new species. *Annals of Carnegie Museum*, 83(4), 311-334. doi:10.2992/007.083.0406

The flea genus *Rectidigitus* [Holland, 1969](#), endemic to Papua New Guinea and Papua Province, Indonesia is reviewed as a continuation of the study of fleas in the Robert Traub flea collection deposited in the Carnegie Museum of Natural History, Pittsburgh, Pennsylvania. This paper (Part IV) is an extension of previous studies by Hastriter ([2012](#)), Hastriter and Easton ([2013](#), Part I, *Striopsylla*), Hastriter ([2014](#), Part II, *Nestivalius*, *Orthopsylloides*, and *Parastivalius*), and Hastriter ([2015](#), Part III, *Traubia*). *Rectidigitus* currently contains four valid species ([Mardon 1981](#)): *R. ancisus* ([Jordan, 1937](#)), *R. spooneri* ([M. Rothschild, 1934](#)), *R. szentivanyi* [Holland, 1969](#), and *R. traubi* [Holland, 1969](#). The male of *R. ancisus* is described for the first time and the previously known distribution of this species is expanded from Morobe Province to Southern Highlands and Western Highlands Provinces. **An additional four new species of *Rectidigitus* are described herein (*R. angularis*, *R. claviculatus*, *R. dittmarae*, and *R. glomerospinosus*).** With the description of these four new species, the total number of described species in the superfamily Pygiopsylloidea in Papua Province, (Indonesia), Papua New Guinea (including Bismarck Archipelago), and the Solomon Islands is 105. An additional eight species belonging to three other flea families (Ichnopsyllidae (3), Pulicidae (3), and Leptopsyllidae (2)) bring the total number of flea taxa to 113 species (including subspecies). A key to the species of *Rectidigitus* is provided.

[*R. angularis*, *R. claviculatus*, *R. dittmarae*, and *R. glomerospinosus* Hastriter 2016]

Hastriter, M. W. (2016). DESCRIPTION OF *WILSONIPSYLLA SPINICOXA*, NEW GENUS AND SPECIES OF FLEA FROM PAPUA NEW GUINEA AND REVIEW OF THE SUBORDER

PYGIOPSYLLOMORPHA (INSECTA: SIPHONAPTERA)(vol 81, pg 19, 2012). ANNALS OF CARNEGIE MUSEUM, 83(4), 337-337.

[*Wilsonipsylla spinicoxa* Hastriter 2016, n. gen., n. sp.]

Hosts and Distribution

Zohdy, S., & Durden, L. A. (2016). 11 A review of ectoparasites in the Cheirogaleidae. The Dwarf and Mouse Lemurs of Madagascar: Biology, Behavior and Conservation Biogeography of the Cheirogaleidae, 73, 220.

Villalobos-Cuevas, V. A., M. Weber, M. Lareschi, R. Acosta. 2016. Pulgas parásitas de mamíferos pequeños y medianos de Calakmul, Campeche, México y nuevos registros de localidades. Revista Mexicana de Biodiversidad, 87: 1372-1378. [Flea parasitizing small and medium mammals of Calakmul, Campeche, México and new records of localities].

Ecology and Evolution

Sponchiado, J., Melo, G. L., Martins, T. F., Krawczak, F. S., Jacinavicius, F. C., Labruna, M. B., ... & Cáceres, N. C. (2016). Ectoparasites of small-mammals: determinants of community structure in South American savannah. Parasitology, 1-9.

This study aimed to assess the contribution of hosts characteristics (rodents and marsupials) in the organization of ectoparasite communities present in woodland patches in western central Brazil. We verified the effect of host species, sex, body mass and vertical strata in addition to the role of seasonality on the ectoparasite composition, richness and abundance.

The total sampling effort was 22 032 trap-nights equally distributed in 54 woodland patches. Variance partition and principal coordinate analysis were used to verify the existence of significant relationships between response variables and predictors.

As expected, host species was the most important variable in ectoparasite community assembly. The composition, richness and abundance of mites and lice were highly influenced by host species, although higher for mites than for lice. Host body mass had a determining role on the richness and abundance of tick species. Vertical stratification and seasonality had weak influence, while the sex of the host had no influence on the organization of these communities. The results are closely related to the evolutionary characteristics of the species involved, as well as with local environmental characteristics of the study area.

Pathology and Control

Chen, Y. J., Huang, C. G., Hsu, J. C., & Wu, W. J. (2016). Development of a Larval Bioassay Method Using 96-Well Microtiter Plates for Evaluation of Susceptibility of the Cat Fleas (Siphonaptera: Pulicidae) to Insecticides. Journal of Medical Entomology 09/2016; DOI:10.1093/jme/tjw156

Linardi, P. M.; de Avelar, D. M. Siphonaptera. In: D. P. Neves; A. L. Melo; P. M. Linardi; R. W. A. Vitor. (Org.). Parasitologia Humana. 13 ed. São Paulo-Rio de Janeiro-Belo Horizonte: Editora

Atheneu, 2016, p. 459-470.

Linardi, P. M. Fleas and Diseases. In: C. B. Marcondes (Ed.). Arthropod Borne Diseases. Springer, 2016, p. 517-536.

Microbial Symbiotes

Campana, M. G., Hawkins, M. T., Henson, L. H., Stewardson, K., Young, H. S., Card, L. R., ... & Helgen, K. M. (2016). Molecular ecology resources. Simultaneous identification of host, ectoparasite and pathogen DNA via in-solution capture. <https://www.ncbi.nlm.nih.gov/pubmed/26990246> doi: [10.1111/1755-0998.12524](https://doi.org/10.1111/1755-0998.12524)

Lučan, R. K., Bandouchova, H., Bartonička, T., Pikula, J., Zahradníková, A., Zukal, J., & Martínková, N. (2016). Ectoparasites may serve as vectors for the white-nose syndrome fungus. *Parasites & vectors*, 9(1), 1. <https://parasitesandvectors.biomedcentral.com/articles/10.1186/s13071-016-1302-2>

“We hypothesise that mechanical transport of *P. destructans* propagules between bats on the bodies of spinturnicid mites is enabled by their specialisation of living on bat wing membranes, i.e., the body region most typically affected by fungal growth [29]. This is supported by our finding of a positive relationship between fungal load on wing mites and fungal load and infection intensity on bat wings. An analogous situation may also hold true for other bat ectoparasites infesting bats during the time of fungal growth, such as Macronyssidae, fleas and nycteribiid flies; however, this remains to be investigated.”

Antonio Zurita, Sara García Gutiérrez, Cristina Cutillas. (2016) Infection Rates of *Wolbachia* sp. and *Bartonella* sp. in Different Populations of Fleas. *Curr Microbiol* 73:704–713. DOI 10.1007/s00284-016-1119-4

In the present study, a molecular detection of *Bartonella* sp. and *Wolbachia* sp. in *Ctenocephalides felis* (Siphonaptera: Pulicidae) isolated from *Canis lupus familiaris* from different geographical areas of Spain, Iran and South Africa, and in *Stenoponia tripectinata tripectinata* isolated from *Mus musculus* from the Canary Islands has been carried out by amplification of the 16S ribosomal RNA partial gene of *Wolbachia* sp. and intergenic spacer region (its region) of *Bartonella* sp. A total of 70 % of *C. felis* analysed were infected by *W. pipientis*. This percentage of prevalence was considerably higher in female fleas than in male fleas. *Bartonella* DNA was not detected in *C. felis* from dogs, while *Bartonella elizabethae* was detected and identified in *S. t. tripectinata* from *M. musculus* from the Canary Islands representing 43.75 % prevalence. This report is the first to identify *B. elizabethae* in *S. t. tripectinata* collected in *M. musculus* from the Canary Islands. Thus, our results demonstrate that this flea is a potential vector of *B. elizabethae* and might play roles in human infection. The zoonotic character of this bartonellosis emphasizes the need to alert public health authorities and the veterinary community of the risk of infection.

Fleas in Art and History

"We have an example of the first (not to speak of spectacles and the like, which only correct and remove the infirmity of a deficient sight, and therefore give no further information) in the lately invented microscopes, which exhibit the latent and invisible minutiae of substances, and their hidden formation and motion, by wonderfully increasing their apparent magnitude. By their assistance we behold with astonishment the accurate form and outline of a flea, moss, and animalculæ, as well as their previously invisible color and motion. It is said, also, that an apparently straight line, drawn with a pen or pencil, is discovered by such a microscope to be very uneven and curved, because neither the motion of the hand, when assisted by a ruler, nor the impression of ink or color, are really regular, although the irregularities are so minute as not to be perceptible without the assistance of the microscope. Men have (as is usual in new and wonderful discoveries) added a superstitious remark, that the microscope sheds a lustre on the works of nature, and dishonor on those of art, which only means that the tissue of nature is much more delicate than that of art. For the microscope is only of use for minute objects, and Democritus, perhaps, if he had seen it, would have exulted in the thought of a means being discovered for seeing his atom, which he affirmed to be entirely invisible. But the inadequacy of these microscopes, for the observation of any but the most minute bodies, and even of those if parts of a larger body, destroys their utility; for if the invention could be extended to greater bodies, or the minute parts of greater bodies, so that a piece of cloth would appear like a net, and the latent minutiae and irregularities of gems, liquids, urine, blood, wounds, and many other things could be rendered visible, the greatest advantage would, without doubt, be derived."

- Francis Bacon (1561-1626), *Novum Organum*, Or True Suggestions for the Interpretation of Nature (1620).

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On the next page

Piazza Mercatello durante la peste a Napoli del 1656 [Piazza Mercatello in Naples during the Plague of 1656] (circa 1675), by Domencio Gargiulo detto Micco Spadaro (b. 1609–1610, Naples, Italy, d. ca. 1675, Naples). Oil on canvas, Museo della Certosa di San Martino, Napoli.



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Siphonaptera Literature (2016)

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